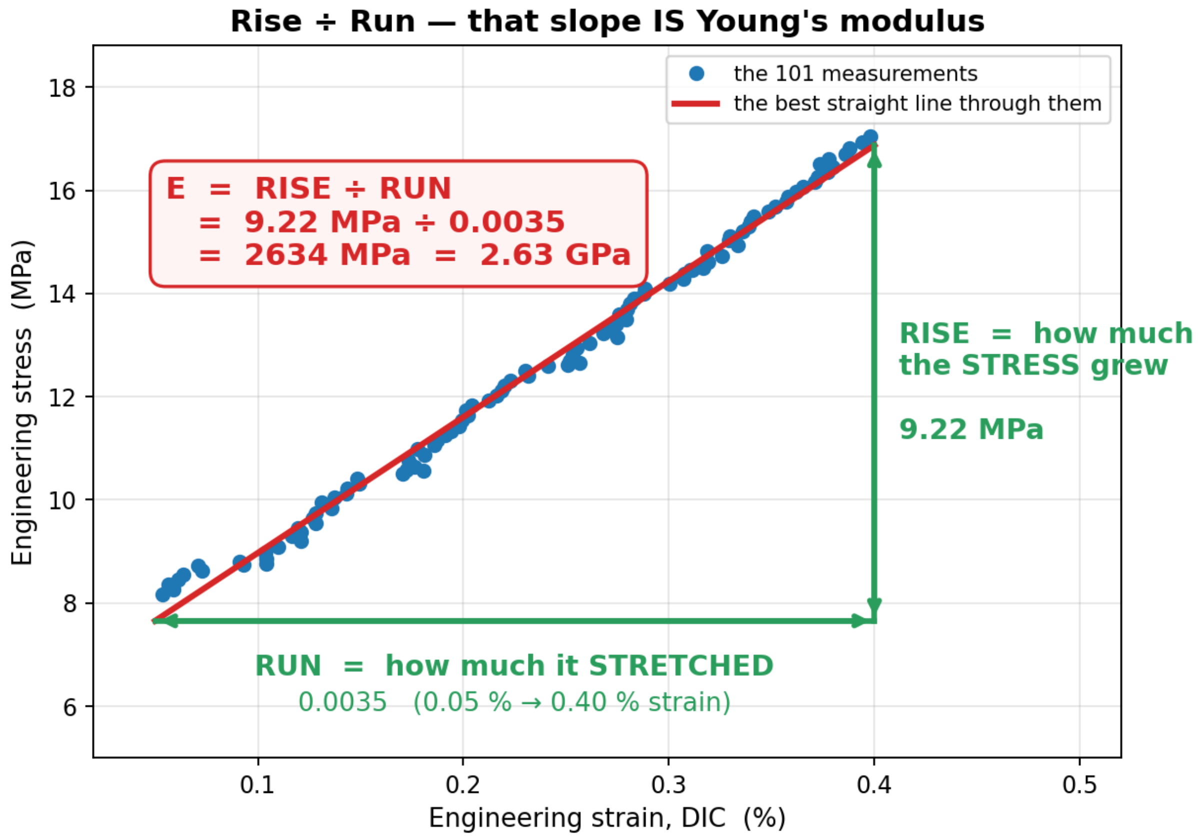
Young's modulus E — how stiff is the material?

Pull the specimen. It stretches. E is how steeply the pull has to rise to keep it stretching.



STRESS = how hard you are pulling ÷ how thick the specimen is. Units MPa.

STRAIN = how much the gauge stretched ÷ how long it was. Just a fraction — no units.

E = how much the STRESS went up, divided by how much it STRETCHED to get there.

```
RISE  =   16.87  -   7.65   =  9.22 MPa
RUN   =  0.0040  - 0.0005  =  0.0035

E  =  9.22 / 0.0035  =  2634 MPa
```

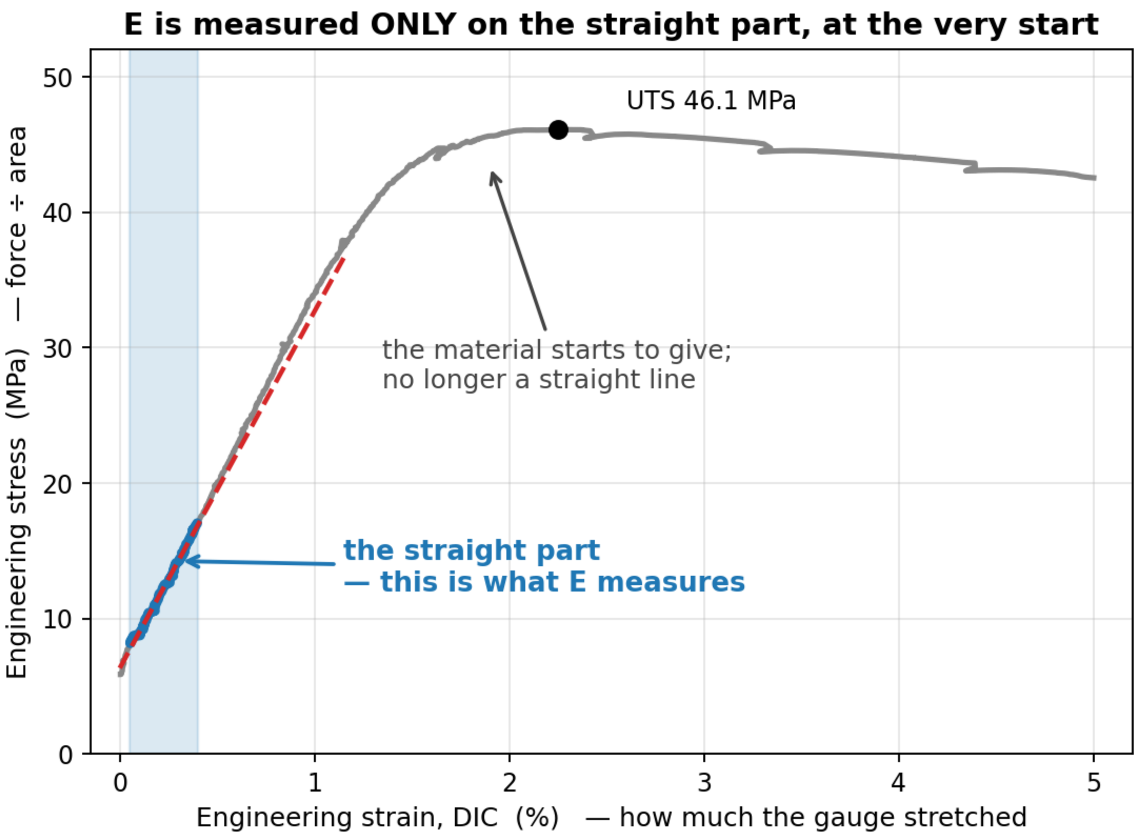
2634 MPa = 2.63 GPa. A stiffer material gives a steeper line and a bigger E; a floppier one gives a shallower line.

Steepness of the straight bit = stiffness. Nothing else on the curve is used for E.

V6d · S11 · 100 % infill — the real measurements, not an illustration

The formula the code actually uses

Two points would do — but 101 were measured, so all 101 are used. That is all "least squares" means.



Best-fit slope through all the points:

$$E = \frac{n \cdot \Sigma(\varepsilon \cdot \sigma) - \Sigma\varepsilon \cdot \Sigma\sigma}{n \cdot \Sigma(\varepsilon \cdot \varepsilon) - (\Sigma\varepsilon)^2}$$

Add up four columns over the 101 points:

```
n          = 101
Σε         = 0.237696
Σσ         = 1266.1435  MPa
Σ(ε·ε)     = 0.000657593
Σ(ε·σ)     = 3.238406  MPa
```

Put the numbers in:

```
top    = 101·3.2384 − 0.2377·1266.14 = 26.1218
bottom = 101·0.0006576 − 0.2377²  = 0.0099175
E = 26.1218 / 0.0099175 = 2634 MPa = 2.63 GPa
```

$\sigma = 2634 \times \varepsilon + 6.34$ $R^2 = 0.9936$ the 6.34 MPa offset is the preload the curve starts from, not an error

utm_analysis.analyze() · fit window 0.05 %–0.40 % strain · σ = (F + 470 N anchor) / 80 mm²